Reiwa 6 (2024) Date submitted: 9/02/2025

Fundamental Labs I Report

(School of Science and Engineering, College of Engineering Systems)

☐ 7 System control engineering basic labs
☐ 8 Basics of linear systems using operational amplifier
☐ 9 Diodes and transistors
☐ 10 Basics of logic circuits and computers
☐ 11 DC motor manufacturing and control
☑ 12 Mechanisms and mechanical elements

Submitted by: Lab group____ Student ID No. 2023B030 Name Andrea Trevanna

Lab partners: (1) Chong Shao (2) _____
(3) _____ (4) _____
(5) _____ (6) _____

	1st week	2nd week	Report submission
Lab date	Month/Day	Month/Day	Received stamp
Lab participant's stamp with seal Non-participants leave blank	基礎実験 25. 1. 27 26-km 機構と機械要素	基礎実験 25. 2. 03 26-km 機構と機械要素	
Latecomer arrival time	:	:	

Use this sheet as the report cover page.
Reports are submitted by the date/time and to the location designated for each theme.
Submission deadline: Month: 02 Date: 10 (), Time: 12: 15, Submission location: Mangla
NOTE: This should be collected and saved by each teaching assistant (TA) after the mandatory information has been initially filled in during the first lab. After their participation in the second lab has been understood, this sheet is returned to students who are instructed to use it as the cover page of their report.

Chapter 12 Kinematics and Mechatronics Report

Introduction

1. Purpose of this lab

• Learn basics on kinematics and mechatronics, such as gear and link mechanisms.

• Assemble actual mechanisms, move them, and learn how the mechanisms work.

• Design and build a mechanism that moves as planned for a given task.

2 Lab summary (a) LEGO MindStorms introduction and programming

(b) Assemble and operate basic mechanical elements and mechanisms

First week: Assignment 1 to Assignment 9

Second week: Assignment 10 to Assignment 17

Week 1

We used LEGO Mindstorms for assembling and operating the various mechanisms and mechanical elements in this lab. LEGO MindStorms consists with various parts (gears, shafts, pins, etc.), motors for driving joints, optical sensors, and touch sensors. Motor and sensors will be connected to EV3 control box. To control the system, we write our own programs using visual programming system, ROBOLAB.

The EV3 control box is connected to the computer via a cable. To run the ROBOLAB program, we click the run button and the program will be transferred/downloaded into the EV3 box.

Assignment 1

1) The program to rotate a motor eternally is below. It consists of start block and a loop box that contains the motor block and rotation command.

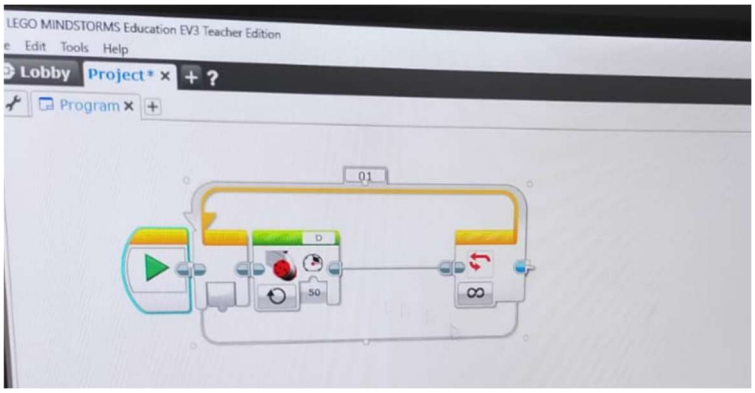


Figure 1 Program code

When we click run, the result is that the motor (red) connected to the EV3 is rotated eternally.



Figure 2 Motor connected to EV3

2) The program to make a motor rotates and stops when you press and release a touch sensor is below

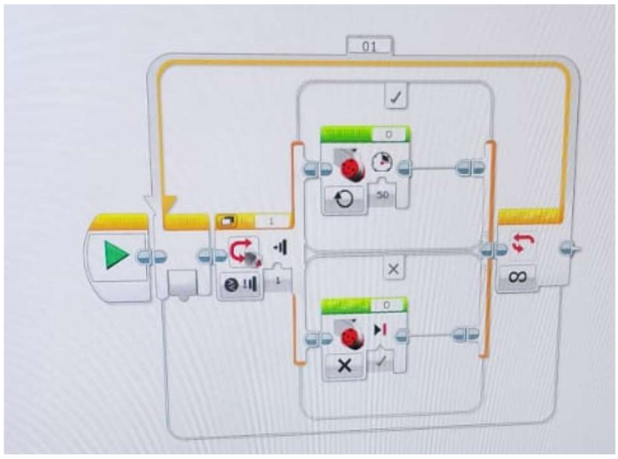


Figure 3 Program code

We connected a touch sensor to the EV3. Then with ROBOLAB, we create the program using start block, and a loop containing a touch sensor and a switch block containing if and else: if the motor is rotating or the motor will not rotate. We click run the program.

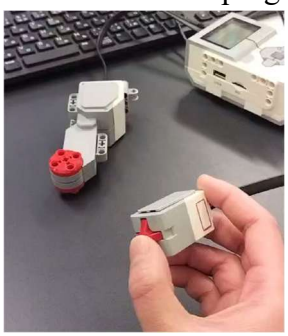


Figure 4 Motor and touch sensor connected to EV3

3) The program to make a motor rotates and stops when you direct an optical sensor to face a bright place and a dark place (Direct the sensor to face a white paper and a black table.) is below

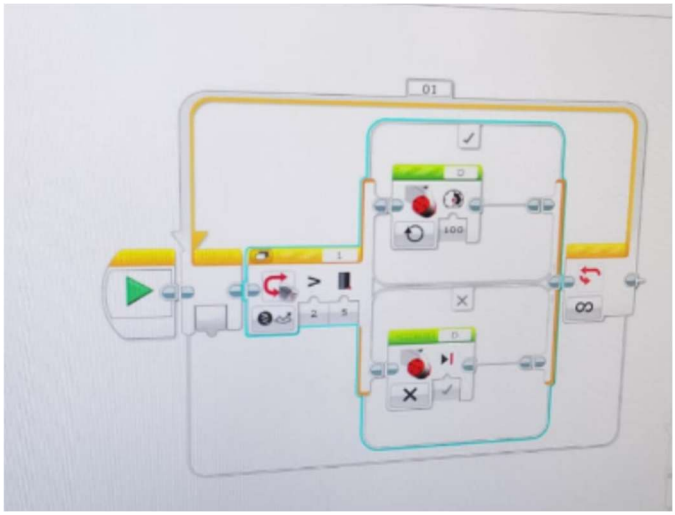


Figure 5 Program code

With the colour sensor connected to EV3 and the colour block inserted to the loop block in the program. The loop block contains the color block connected to the switch block of the motor which contains two conditions: motor rotates or the motor does not rotate.



Figure 6 Motor and color sensor connected to EV3

Assignment 2

Using the touch sensor and motor, we created the program so that when you press touch sensor, it will make reverse rotations.

The program is below



Figure 7 Program code

Assignment 3

Using the distance sensor and motor, we created the program so that when the distance with the wall is 1) near, the motor will rotate forward, 2) middle distance, the motor will stop, 3) far, the motor will rotate backwards. We connected the distance sensor to the EV3. Then we created the program using distance sensor block connected to a switch block containing different distances and how the motor should act (if-else).

The program is below.

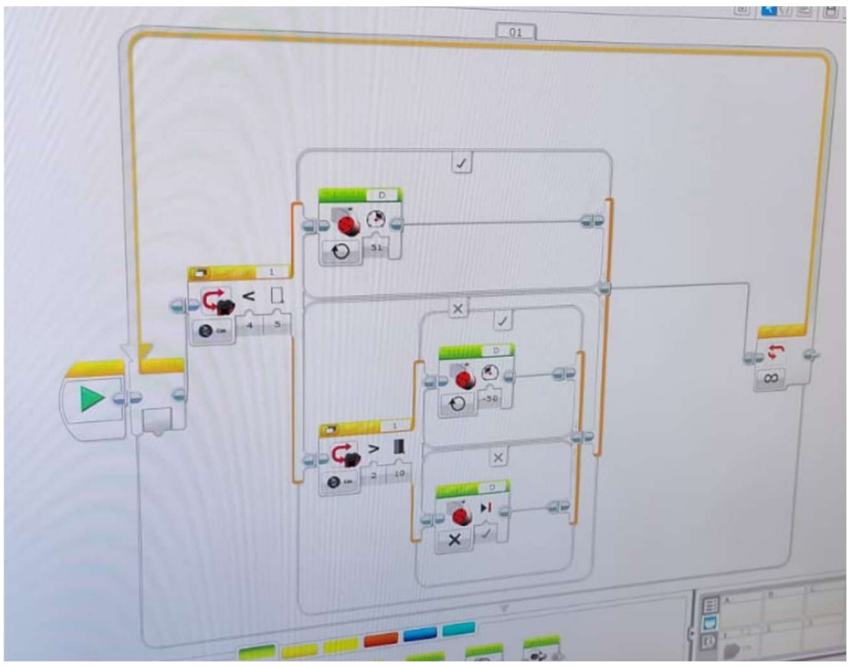


Figure 8 Program code

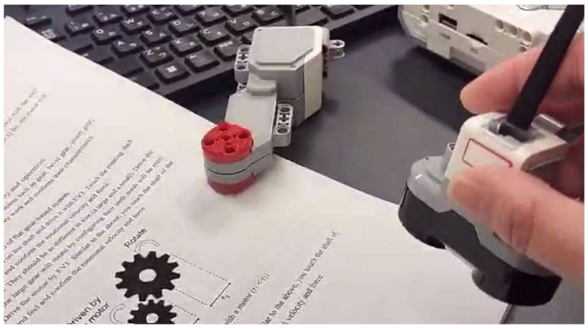


Figure 9 Using distance sensor attached to EV3 to rotate the motor given distance conditions

Assignment 4

We prepared two kinds of flat gears. They have two different sizes(large and a small). We then did a different experiments to the gears, respectively :

1) We drove the small gear with a motor then the large gear will rotate by configuring their teeth mesh will be meet with each other (Fig. 14). Drive the motor by EV3. We touched the shaft of the large gear with our finger and felt and confirmed the rotational velocity and force.

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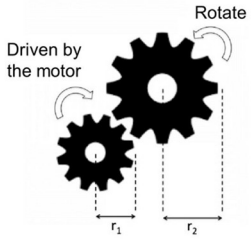


Figure 10 Drive the small gear with a motor (r1<r2)

2) In next, drove the large gear by the motor with EV3. Similar to the above, we touched the shaft of the large gear with our finger and felt and confirmed the rotational velocity and force.

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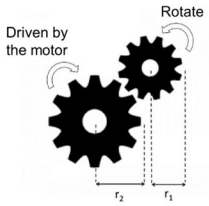
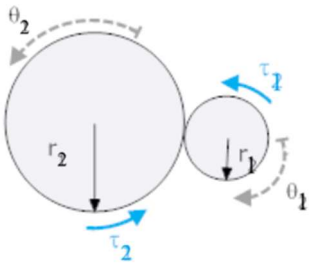


Figure 11 Drive the large gear with motor (r1<r2)

Based on these experiments, we obtained the following result:

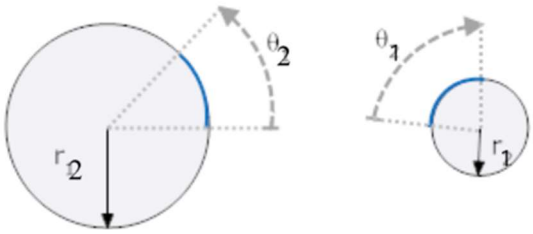
The smaller gear (with radius r_1) has faster rotation and more force (stronger sensation on the finger when the finger touched the shaft), while the larger gear (with radius r_2) has slower rotation and less force.

The mathematical relation of the rotational velocity and the two gears can be explained using the following formula using r_1, r_2, torque τ_1 of the smaller gear and torque τ_2 of the larger gear, and rotational velocity ω_1 of the smaller gear and rotational velocity ω_2 of the larger gear:

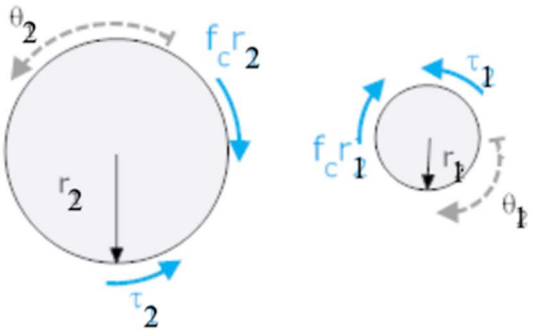


If we consider two gears in equilibrium and in contact with each other, we can obtain these two relationships:

1) the geometric relationship that results from the path that the arc lengths along their circumference must be equal as the gears turn.



2) We derive a second relationship from a torque balance. Before we can do so we must define a force between the gears termed a "contact force.", f_c . This force must be equal and opposite across the interface between the two gears, but its direction is arbitrary. We start by drawing free body diagrams with a contact force where the gears meet.



The contact force is tangent to both gears and so produces a torque that is equal to the radius times the force.

Gear 1(Small gear) : Gear 2(Big Gear):

$$\tau_1 = f_c r_1 \quad \text{so } f_c = \frac{\tau_1}{r_1} \qquad \tau_2 = f_c r_2 \quad \text{so } f_c = \frac{\tau_2}{r_2}$$

We are not usually interested in f_c, so we remove from the equations and we get

$$\frac{\tau_1}{r_1} = \frac{\tau_2}{r_2}$$
$$\tau_1 r_2 = \tau_2 r_1$$

Based on this relation, we obtain $\tau_2 = \frac{\tau_1}{r_1} r_2$. This is the reduction ratio.

If we combine the geometric relationship with the torque relationship

$$\theta_1 = \frac{r_2}{r_1}\theta_2, \quad \text{or } \omega_1 = \frac{r_2}{r_1}\omega_2$$

$$\tau_1 = \frac{r_1}{r_2}\tau_2$$

the product of torque and angular velocity becomes:

$$\tau_1\omega_1 = \left(\frac{r_1}{r_2}\tau_2\right)\left(\frac{r_2}{r_1}\omega_2\right) = \tau_2\omega_2$$

The product of angular velocity and torque is power. So, the last equation simply states that power is preserved across a set of gears. The gears allow one to trade torque for angular velocity (i.e., if the gears are of unequal size, the larger gear will experience higher torque but lower angular velocity than the smaller gear).

Gears and Systems with both Rotation and Translation
https://lpsa.swarthmore.edu/Systems/MechRotating/RotMechSysGears.html

Assignment 5

We prepared three gears: small with radius r_1 , medium with radius r_2 , and large with radius r_3. ($r_1 < r_2 < r_3$) Drive the small gear with a motor, then make its teeth mesh

will be met to the medium gear and then, the teeth of the medium gear mesh will meet to the large gear. Rotate the motor and confirm the shaft rotational velocity and force of the large gear when they are rotating.

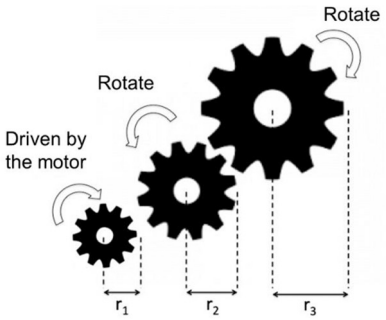


Figure 12 Engaging large, medium and small gears (r1 < r2 < r3)

The smallest gear: Quickest rotation, strongest force

Medium gear: Medium rotation, medium force

Largest gear: Slowest rotation, more force

The mathematical relation to show the rotational velocity and force is below:

From the relation in Assignment 4, we can write:

$$f_c = \frac{\tau_1}{r_1} = \frac{\tau_2}{r_2} = \frac{\tau_3}{r_3}$$

The deceleration ratio is $\tau_3 = \frac{\tau_1}{r_1} r_3$, which is independent of the medium gear

With $\omega = \frac{f}{r}$, $\omega_1 = \frac{r_3}{r_1}\omega_3$

Assignment 6:

See Assignment 5 for the deceleration ratio $\tau_3 = \frac{\tau_1}{r_1} r_3$

The deceleration ratio is constant as it only depends on the torques and ratios of gear 1 and gear 3. Thus, the value of the medium gear is independent of the deceleration ratio and the deceleration ratio is always constant. Thus, the only way we can get very large the deceleration ratio is if the small gear has very small radius and the large gear has very large radius, which is not so efficient or easy.

To achieve a very large deceleration ratio using multiple flat gears, a compound gear train mechanism can be employed. This mechanism uses several pairs of gears in series to multiply the overall deceleration ratio. In a compound gear train:

1. Multiple gear pairs are connected in series.

2. The output shaft of one gear pair becomes the input shaft for the next pair.
3. The overall velocity ratio (deceleration ratio) is the product of individual gear pair ratios.

Assignment 7: What is backlash?

Gear backlash is a mechanical engineering term referring to the lost motion resulting from spaces occurring between gear teeth during the mesh process of bidirectional gears. For smooth rotation of meshed gears, backlash is necessary. Backlash is the amount by which a tooth space exceeds the thickness of a gear tooth engaged in mesh. Backlashes are classified in the following ways.

(1) Circumferential Backlash (j t)
Circumferential Backlash is the length of arc on the pitch circle. The length is the distance the gear is rotated until the meshed tooth flank makes contacts while the other mating gear is held stationary.
(2) Normal Backlash (j n)
The minimum distance between each meshed tooth flank in a pair of gears, when it is set so the tooth surfaces are in contact.
(3) Angular Backlash (j θ)
The maximum angle that allows the gear to move when the other mating gear is held stationary.
(4) Radial backlash (j r)
The radial Backlash is the shrinkage (displacement) in the stated center distance when it is set so the meshed tooth flanks of the paired gears get contact each other.
(5) Axial Backlash (j x)
The axial backlash is the shrinkage (displacement) in the stated center distance when a pair of bevel gears is set so the meshed tooth flanks of the paired gears contact each other.

Sources for Assignment 7:

1. Gear Backlash | KHK Gear Manufacturer

https://khkgears.net/new/gear_knowledge/gear_technical_reference/gear_backlash.html

2. Gear Backlash Guide | WM Berg
https://www.wmberg.com/resources/blogs/guide-to-gear-mesh-transmission-backlash

Assignment 8

To reverse the rotation directions, there are two methods 1) using bevel gears and 2) using worm gears. First, we assembled a mechanism consisting of a motor, bevel gear,

and worm gear. We then rotated the motor by EV3. The two wheels will rotate through the worm gear and the bevel gear.

1) a. The deceleration ratio for the bevel gear is the same as before:

$$\omega_2 = \frac{r_1}{r_3}\omega_1$$

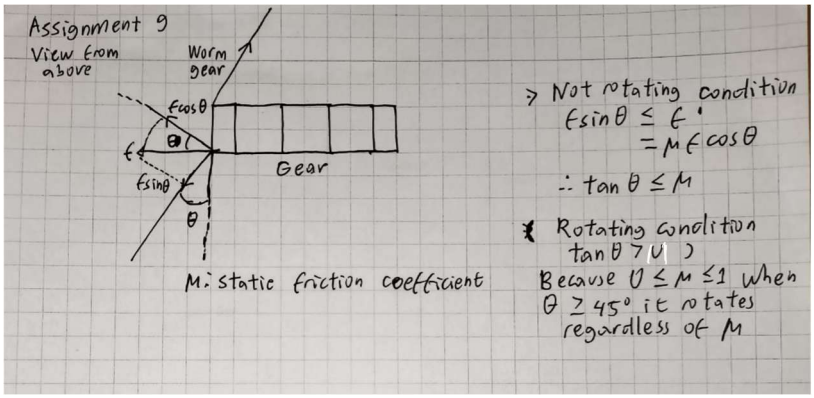
b. The deceleration ratio for the worm gear is

$\omega_2 = \frac{1}{n}\omega_1$, where n is the number of teeth of the worm gear

Assignment 9

Although we may rotate the worm gear by rotating wheel 1, we may not be able to rotate the worm gear by rotating wheel 2.

The reason is explained by the following diagram,



Week 2

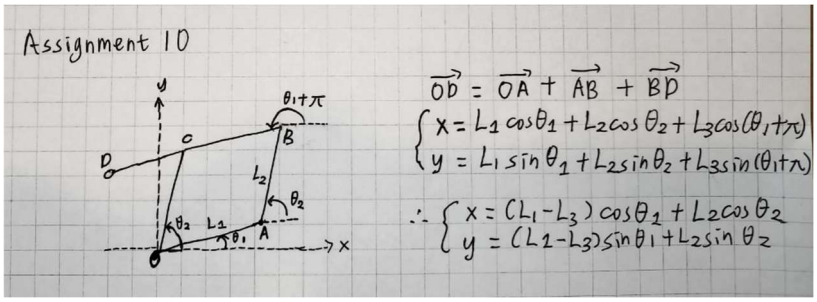
We assembled the parallel four-link mechanism below ,which is a typical linkage mechanism.



Figure 13 Parallel four-link mechanism

Assignment 10

The coordinates of the hands of the parallel four links



Assignment 11 (attached to the report)

We recorded the tip operation of the parallel four links on graph paper (A3) as in Figure 14. First, we set all angles to 90° (turning the parallel four-sided shape into a rectangle), recorded the path when rotating only the top gear from this orientation, and the path when rotating only the bottom gear, and discuss the mechanism characteristics. In addition, we recorded the overall range of movement.

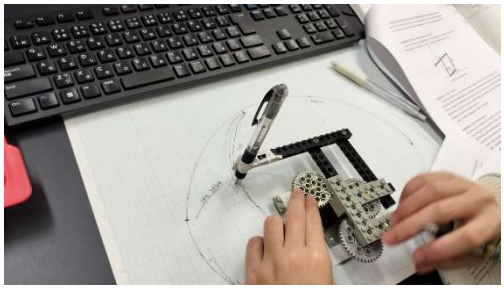


Figure 14 Recording tip operation of the parallel four links using the tip of a pen attached to the four link

The graph paper of the paths is attached to the report as Graph 1. The labelling of gear A (bottom gear) and gear B(top gear) in Graph 1 is according to the figure below

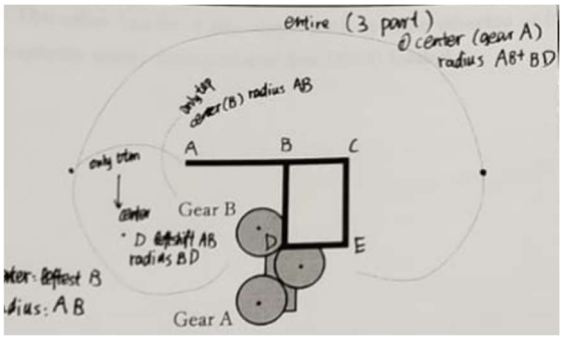


Figure 15 Labelling of Graph 1 and the parallel four link

Assignment 12

The advantages of the parallel four-link mechanism when the motor is placed on a base that is not the link joint vs the disadvantages when the motor is placed on the linkage:

1) Weight Distribution

Base: Reduces stress on linkage; increases performance
Linkage: shape change, structure unstable

2) Size:

Base: Size has lower effect on link movement

Linkage: Size of motor may prevent motion

3) Power supply:
Base: fixed length of wiring
Linkage: Wiring is harder (motor may moving), risk of act off

We then modified the link length so the parallel four link mechanism became "non-parallel" four link mechanism.

Assignment 13

1. Disadvantage of the non-parallel four link:
Shorter movement than parallel
2. Have to calculate extra angle if the lines are not parallel

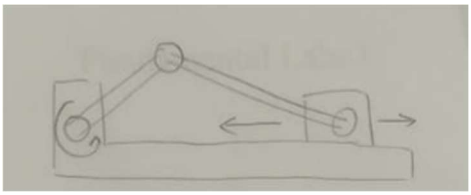
Advantage:

Parallel is simpler than non parallel, since the angles are 90 degrees apart (sin 90° =1)

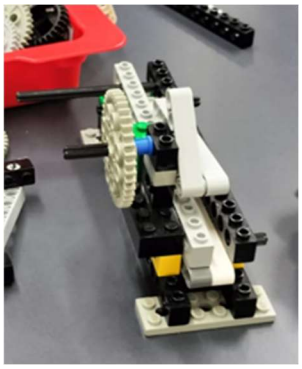
Assignment 14

We created these three mechanisms for converting the rotational motion of a motor into reciprocal motion using Lego.

1) The crank-and-slider mechanism: The rotation of the crank drives the linear movement of the slider
Diagram:

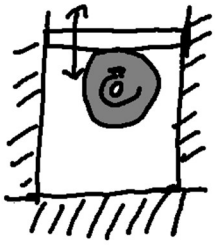


Final:

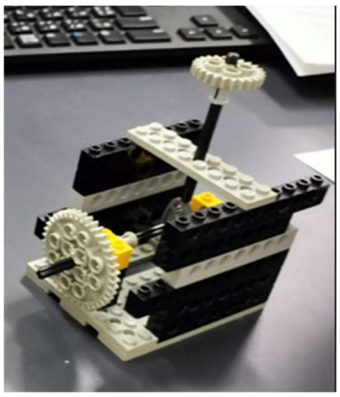


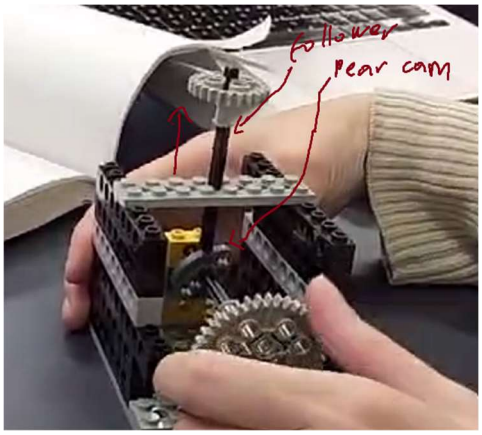
2) Cam mechanism: The cam is the part of the mechanism that rotates. The outer edge of the cam is in contact with a follower. As the cam rotates, it pushes the follower. The movement of the cam causes the follower to move in a direction that is different to that of the cam.

   Diagram:



   Final:





   As the cam is rotated, the cam pushes the follower (the black shaft) upwards

3) Linkage mechanism, which can be rotated in all directions



When rotated becomes



Sources for Assignment 4:

1) For Cam Mechanism
a) Chapter 6. Cams
https://www.cs.cmu.edu/~rapidproto/mechanisms/chpt6.html
b) What is a cam mechanism? - KS2 - BBC Bitesize
https://www.bbc.co.uk/bitesize/articles/z2gfcxs#zv4s46f
2) For Crank and Slider mechanism:
Wikipedia Slider Crank Linkage
https://en.wikipedia.org/wiki/Slider-crank_linkage

We then made a Paucellier exact straight-line motion mechanism like below for assignments 15 and 16

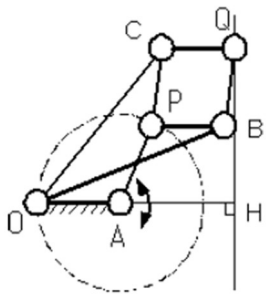
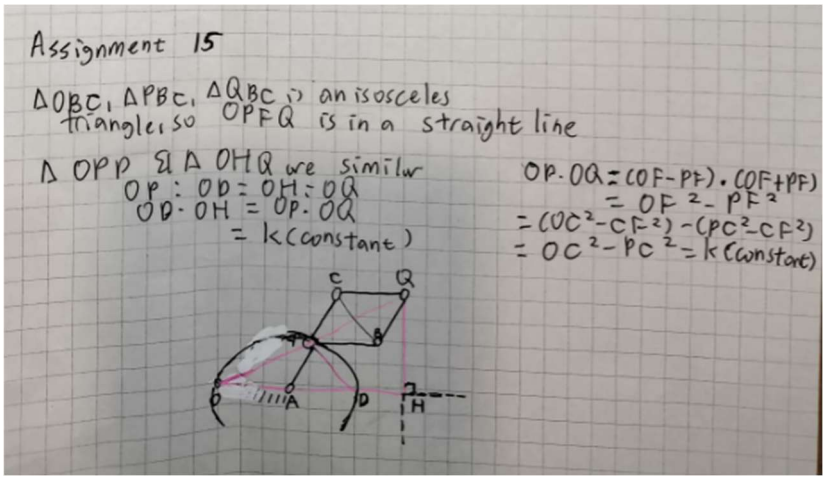


Figure 16 Exact straight-line motion

Assignment 15

Prove geometrically that Q is a straight line



Assignment 16

We recorded the path of point Q on graph paper using the tip of the pen attached to the linkage mechanism (similar to Assignment 11), and confirmed the range of the operation as in Figure 16.

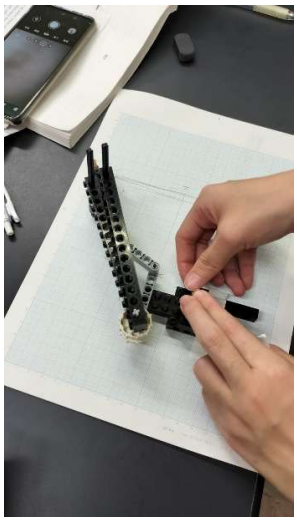


Figure 17 Doing Assignment 16 similar to Assignment 11 (Figure 14) to record the tip of the linkage as it does the exact straight-line motion mechanism

The graph paper is attached to the report as Graph 2

Assignment 17

For this assignment, we used LEGO Mindstorms. We used two sensors: touch sensor (detects contact and non-contact with objects) and color sensor (block (with emits LED light, and then determines whether objects are present or are black or white according to

the intensity of the reflected light) , the motors, and the tires, and built a mobile robot for that keeps moving back and forth between walls by recognizing walls with sensors. We incorporated EV3 into the robot so that the program we write can be inserted into the robot.

How we built the robot:

First, we attached motors and 4 tires into the EV3 box. Then, we connected the touch and color sensors to the EV3. From Figures 20 and 21, we can see that the touch sensor is in the back and the color sensor is in the front side of the robot. The color sensor acts as the robot's "eye" by detecting the intensity of the light (light and dark spaces).

We wrote the following program in Figure 18, using two loops inside a loop flow block for the continuity of the program when implemented in the robot, using the motor block, color sensor block (with the intensity of the reflected light criteria), and touch sensor. We added the command to move forwards and backwards.

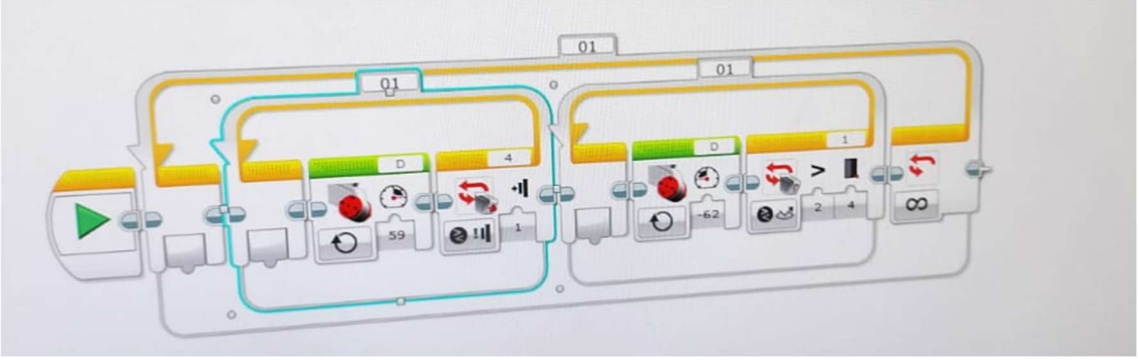


Figure 18 Program code

The robot is in the image below. It moved forward to the wall and then move backward when we pressed the touch sensor.



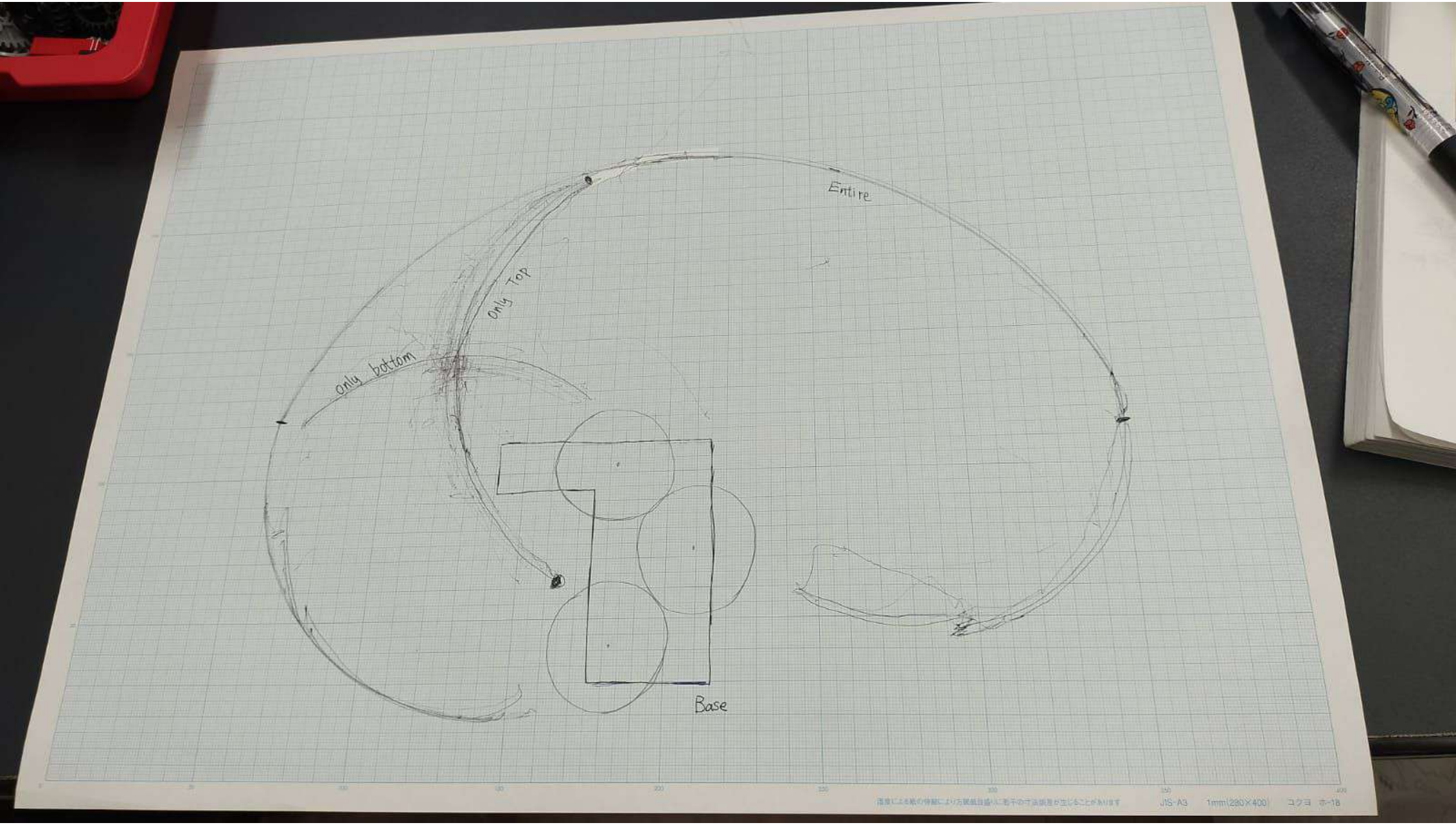
Figure 19 The robot from side view as it hit the wall



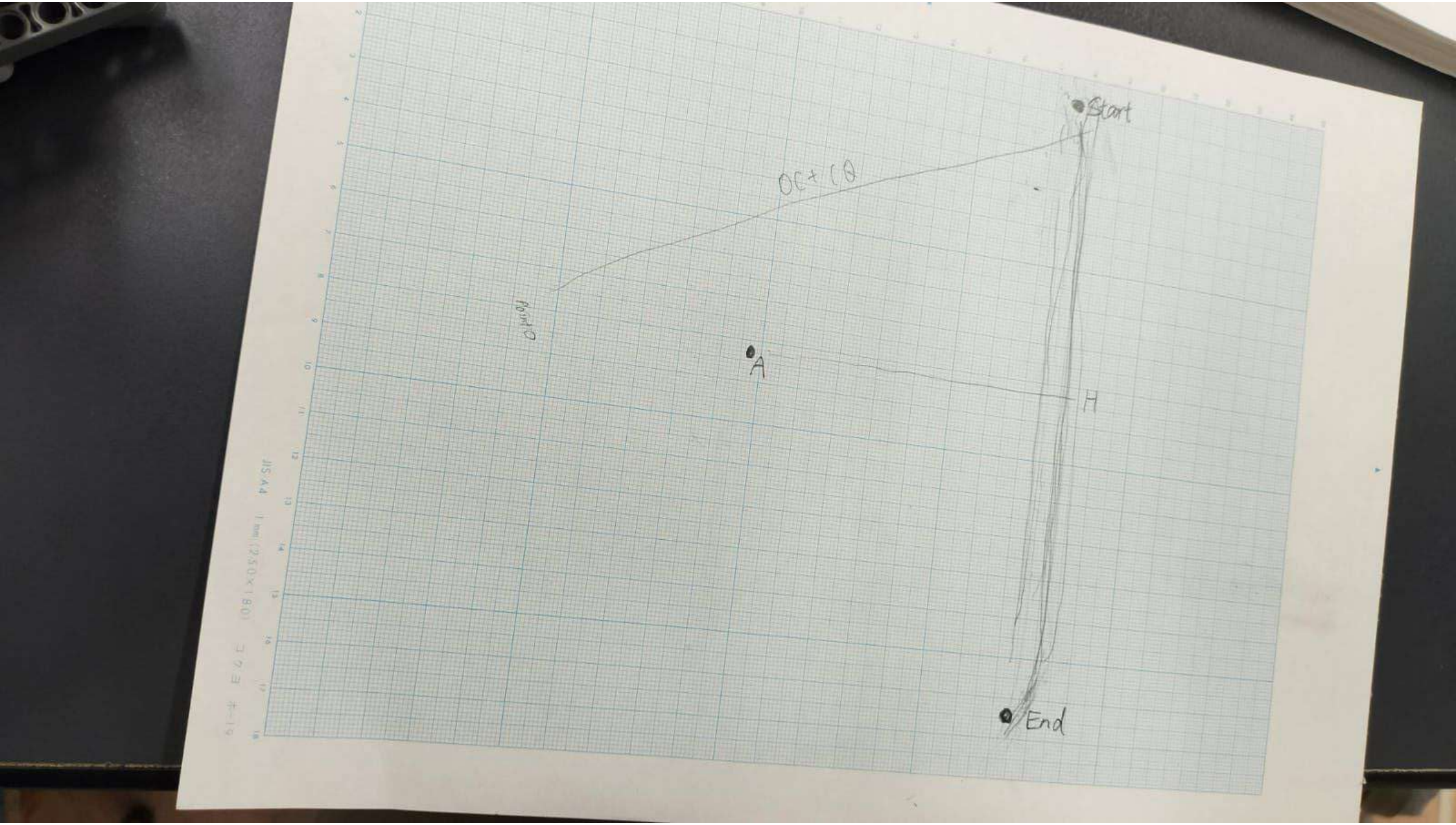
Figure 20 The touch sensor is connected to the back side of the robot



Figure 21 The color sensor is connected to the front side of the robot



Graph 1 Assignment 11 The tip operation of the parallel four links on graph paper (A3)



Graph 2 Assignment 16